1) 1.1)

a - $\begin{pmatrix} 1 & 2 \\ 3 & 2 \end{pmatrix} = 1 \times 2 - 2 \times 3 = 2 - 6 = -4$

b - $\begin{pmatrix} 1 & i \\ i & -1 \end{pmatrix} = 1 \times (-1) - i \times i = -1 - i^2 = 0$

c - $\begin{vmatrix} 0 & 3 & 1 \\ 3 & 2 & 4 \\ 1 & 1 & 2 \end{vmatrix} = 0 \times 2 \times 2 + 1 \times 3 \times 1 + 4 \times 3 \times 1$
$- 1 \times 2 \times 1 - 0 \times 1 \times 4 - 3 \times 3 \times 2$

$= -5$

d - $\begin{pmatrix} 1 & 1 & -1 \\ 2 & -1 & 1 \\ 1 & 4 & 1 \end{pmatrix} = 1 \times (-1) \times 1 + (-1) \times 2 \times 4 + 1 \times 1 \times 1$
$- (-1) \times (-1) \times 1 - 1 \times 4 \times 1 - 2 \times 1 \times 1$

$= -15$

e - $\begin{pmatrix} 3 & 0 & 0 & 0 \\ 1 & 2 & 0 & 0 \\ 2 & 4 & 8 & 0 \\ 1 & 2 & 1 & 1 \end{pmatrix} = 3 \times \begin{vmatrix} 2 & 0 & 0 \\ 4 & 8 & 0 \\ 2 & 1 & 1 \end{vmatrix}$

$= 3 \times (2 \times 8 \times 1 + 0 \times 4 \times 1 + 0 \times 0 \times 2 - 0 \times 8 \times 2$
$- 2 \times 1 \times 0 - 4 \times 1 \times 0)$

$= 3 \times 16 = 48$

1.2) a - $\begin{pmatrix} 1 & 1 & 0 \\ -1 & 2 & 1 \\ 2 & 3 & 1 \end{pmatrix}$

i) $\det(a) = 1 \times 2 \times 1 + 0 \times (-1) \times 3 + 1 \times 1 \times 2 - 0 \times 2 \times 2$
$- 1 \times 3 \times 1 - (-1) \times 1 \times 1$

$= 2$

ii) $\det(a) = 1 \times \begin{vmatrix} 2 & 1 \\ 3 & 1 \end{vmatrix} - 1 \begin{vmatrix} -1 & 1 \\ 2 & 1 \end{vmatrix} + 0 \times \begin{vmatrix} -1 & 2 \\ 2 & 3 \end{vmatrix}$

$= 1 \times (2 \times 1 - 1 \times 3) - 1(-1 \times 1 - 1 \times 2) + 0$

$= -1 + 3 = 2$

$b = \begin{vmatrix} 1 & -3 & 0 & 0 \\ 5 & -2 & 2 & 1 \\ -2 & 4 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{vmatrix}$

i) $\det(b) = 1 \times \begin{vmatrix} -2 & 2 & 1 \\ 4 & 0 & 1 \\ 0 & 0 & 1 \end{vmatrix} + 3 \times \begin{vmatrix} 5 & 2 & 1 \\ -2 & 0 & 1 \\ 0 & 0 & 1 \end{vmatrix} + 0 - 0$

$= 1 \times (-8) + 3 \times 4 = 4$

$\begin{vmatrix} -2 & 2 & 1 \\ 4 & 0 & 1 \\ 0 & 0 & 1 \end{vmatrix} = -2 \times 0 \times 1 + 1 \times 4 \times 0 + 1 \times (-2) \times 0 - 1 \times 0 \times 0$
$- (-2) \times 0 \times 1 - 4 \times 1 \times 2$

$= -8$

$\begin{vmatrix} 5 & 2 & 1 \\ -2 & 0 & 1 \\ 0 & 0 & 1 \end{vmatrix} = 5 \times 0 \times 1 + 1 \times 0 \times (-2) + 1 \times 0 \times 2 - 1 \times 0 \times 0$
$- 5 \times 0 \times 1 - (-2) \times 1 \times 2$

$= 4$

$$c - \begin{pmatrix} 1 & 0 & 1 & 1 \\ -1 & 1 & 2 & -1 \\ 1 & 0 & 2 & 0 \\ 2 & 3 & -2 & 4 \end{pmatrix} = 1 \times \begin{vmatrix} 1 & 2 & -1 \\ 0 & 2 & 0 \\ 3 & -2 & 4 \end{vmatrix} - 0 \times \begin{vmatrix} -1 & 2 & -1 \\ 1 & 2 & 0 \\ 2 & -2 & 4 \end{vmatrix}$$

$$+ 1 \times \begin{vmatrix} -1 & 1 & -1 \\ 1 & 0 & 0 \\ 2 & 3 & 4 \end{vmatrix} - 1 \begin{vmatrix} -1 & 1 & 2 \\ 1 & 0 & 2 \\ 2 & 3 & -2 \end{vmatrix}$$

$$= 1 \times 14 - 0 + 1 \times (-7) - 1 \times 18 = -11$$

$$\begin{vmatrix} 1 & 2 & -1 \\ 0 & 2 & 0 \\ 3 & -2 & 4 \end{vmatrix} = 1 \times 2 \times 4 + (-1) \times (-2) \times 0 + 0 \times 3 \times 2 - (-1) \times 2 \times 3$$
$$- 1 \times (-2) \times 0 - 0 \times 2 \times 4$$

$$= 14$$

$$\begin{vmatrix} -1 & 1 & -1 \\ 1 & 0 & 0 \\ 2 & 3 & 4 \end{vmatrix} = -1 \times 0 \times 4 + (-1) \times 3 \times 1 + 0 \times 1 \times 2 - (-1) \times 0 \times 2$$
$$- (-1) \times 3 \times 0 - 1 \times 1 \times 4$$

$$= -7$$

$$\begin{vmatrix} -1 & 1 & 2 \\ 1 & 0 & 2 \\ 2 & 3 & -2 \end{vmatrix} = -1 \times 0 \times (-2) + 2 \times 3 \times 1 + 2 \times 1 \times 2 - 2 \times 0 \times 2 - (-1) \times 3 \times 2$$
$$- 1 \times 1 \times (-2)$$

$$= 18$$

3.3) $\det(A) = aei + chd + fgb - ceg - ahf - dib$

a -

$$3A = \begin{pmatrix} 3a & 3b & 3c \\ 3d & 3e & 3f \\ 3g & 3h & 3i \end{pmatrix}$$

$$\det(3A) = 3a \times 3a \times 3i + 3c \times 3h \times 3d + 3f \times 3g \times 3b - 3c \times 3e \times 3g - 3a \times 3h \times 3f - 3d \times 3i \times 3b$$
$$= 3^3(\det(A)) = 27\, de(A) = 27 \times 2 = 54$$

b- $\begin{vmatrix} -a & -b & -c \\ 2g & 2h & 2i \\ 3d & 3e & 3f \end{vmatrix}$

$= -1\times 2\times 3\, ahf + (-1)\times 3\times 2\, ceg + 2\times 3\times(-1)\, iea - (-1)\times 2\times 3\, chd - (1)\times 3\times 2\, aei$

$- 2\times 3\times(-1)\, gec$

???

$= -6\,(\det(A)) = -6\times 2 = -12$

c- 2

d- $\begin{vmatrix} a & b & c \\ a-d & b-e & c-f \\ g & h & i \end{vmatrix} = \underbrace{\begin{vmatrix} a & b & c \\ a & b & c \\ g & h & i \end{vmatrix}}_{0} - \begin{vmatrix} a & b & c \\ d & e & c \\ g & h & i \end{vmatrix}$

$= -\det(A) = -2$

3.4) a- $\begin{pmatrix} 1 & 3 & 2 \\ -2 & -4 & 0 \\ 3 & 0 & -9 \end{pmatrix} \xrightarrow[L_3 \leftarrow L_3 - 3L_1]{OL_1} \begin{pmatrix} 1 & 3 & 2 \\ -2 & -4 & 0 \\ 0 & -9 & -15 \end{pmatrix}$

$\xrightarrow[L_2 \leftarrow L_2 + 2L_1]{OL_2} \begin{pmatrix} 1 & 3 & 2 \\ 0 & 2 & 4 \\ 0 & -9 & -15 \end{pmatrix} \xrightarrow[L_3 \leftarrow L_3 + \frac{9}{2} L_2]{OL_3} \begin{pmatrix} 1 & 3 & 2 \\ 0 & 2 & 4 \\ 0 & 0 & 3 \end{pmatrix}$

$\begin{vmatrix} 1 & 3 & 2 \\ 0 & 2 & 4 \\ 0 & 0 & 3 \end{vmatrix} = 1\times 2\times 3 = 6$

b - $\begin{pmatrix} 1 & 2 & -1 & 3 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 4 & -1 \\ 1 & 0 & 2 & 4 \end{pmatrix} \xrightarrow[L_4 \leftarrow L_4 - L_1]{OL_1} \begin{pmatrix} 1 & 2 & -1 & 3 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 4 & -1 \\ 0 & -2 & 3 & 1 \end{pmatrix}$

$\xrightarrow[\substack{L_3 \leftarrow L_3 - L_2 \\ L_4 \leftarrow L_4 + 2L_2}]{OL_2} \begin{pmatrix} 1 & 2 & -1 & 3 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 4 & -2 \\ 0 & 0 & 3 & 3 \end{pmatrix} \xrightarrow[L_4 \leftarrow L_4 - \frac{3}{4} L_3]{OL_3} \begin{pmatrix} 1 & 2 & -1 & 3 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 4 & -2 \\ 0 & 0 & 0 & 9/2 \end{pmatrix}$

$$\begin{vmatrix} 1 & 2 & -1 & 3 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 4 & -2 \\ 0 & 0 & 0 & 9/2 \end{vmatrix} = (-1)^{1+1} \times 1 \times \begin{vmatrix} 1 & 0 & 1 \\ 0 & 4 & -2 \\ 0 & 0 & 9/2 \end{vmatrix} + (-1)^{1+2} \begin{vmatrix} 0 & 0 & 1 \\ 0 & 4 & -2 \\ 0 & 0 & 9/2 \end{vmatrix} \times 2$$

$$+ (-1)^{1+3} \times \begin{vmatrix} 0 & 1 & 1 \\ 0 & 0 & -2 \\ 0 & 0 & 9/2 \end{vmatrix} \times (-1) + (-1)^{1+4} \times 3 \times \begin{vmatrix} 0 & 1 & 0 \\ 0 & 0 & 4 \\ 0 & 0 & 0 \end{vmatrix}$$

$$= 1 \times 4 \times \frac{9}{2} = 18$$

c - $\begin{pmatrix} 0 & 0 & 2 & 1 & 1 \\ 0 & 1 & -1 & 3 & 0 \\ 0 & -1 & 0 & 2 & 0 \\ 1 & 2 & 0 & 6 & -3 \\ 0 & 0 & 0 & 1 & 1 \end{pmatrix} \xrightarrow[L_4 \leftrightarrow L_1]{OL_1} \begin{pmatrix} 1 & 2 & 0 & 6 & -3 \\ 0 & 1 & -1 & 3 & 0 \\ 0 & -1 & 0 & 2 & 0 \\ 0 & 0 & 2 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{pmatrix}$

$\xrightarrow[L_3 \leftarrow L_3 + L_2]{OL_2} \begin{pmatrix} 1 & 2 & 0 & 6 & -3 \\ 0 & 1 & -1 & 3 & 0 \\ 0 & 0 & -1 & 5 & 0 \\ 0 & 0 & 2 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{pmatrix} \xrightarrow[L_4 \leftarrow L_4 + 2L_3]{OL_3} \begin{pmatrix} 1 & 2 & 0 & 6 & -3 \\ 0 & 1 & -1 & 3 & 0 \\ 0 & 0 & -1 & 5 & 0 \\ 0 & 0 & 0 & 11 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{pmatrix}$

$$\xrightarrow[L_4 \leftrightarrow L_5]{OL_4} \begin{pmatrix} 1 & 2 & 0 & 6 & -3 \\ 0 & 1 & -1 & 3 & 0 \\ 0 & 0 & -1 & 5 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 11 & 1 \end{pmatrix} \xrightarrow[L_5 \leftarrow L_5 - 11L_4]{OL_5} \begin{pmatrix} 1 & 2 & 0 & 6 & -3 \\ 0 & 1 & -1 & 3 & 0 \\ 0 & 0 & -1 & 5 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & -10 \end{pmatrix}$$

$$\begin{vmatrix} 1 & 2 & 0 & 6 & -3 \\ 0 & 1 & -1 & 3 & 0 \\ 0 & 0 & -1 & 5 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & -10 \end{vmatrix} = 1 \times 1 \times (-1) \times 1 \times (-10)$$
$$= 10$$

3.5) $$A_K = \begin{pmatrix} 1 & 0 & -1 & 0 \\ 2 & -1 & -1 & K \\ 0 & K & -K & K \\ -1 & 1 & 1 & 2 \end{pmatrix} \xrightarrow[L_1 \leftrightarrow L_4]{OL_1} \begin{pmatrix} -1 & 1 & 1 & 2 \\ 2 & -1 & -1 & K \\ 0 & K & -K & K \\ 1 & 0 & -1 & 0 \end{pmatrix}$$

$$\xrightarrow[L_4 \leftarrow L_4 + L_1]{OL_2} \begin{pmatrix} -1 & 1 & 1 & 2 \\ 2 & -1 & -1 & K \\ 0 & K & -K & K \\ 0 & 1 & 0 & 2 \end{pmatrix} \xrightarrow[L_2 \leftarrow L_2 + 2L_1]{OL_3} \begin{pmatrix} -1 & 1 & 1 & 2 \\ 0 & 1 & 1 & K+4 \\ 0 & K & -K & K \\ 0 & 1 & 0 & 2 \end{pmatrix}$$

$$\xrightarrow[L_3 \leftarrow L_3 - K L_2]{OL_4} \begin{pmatrix} -1 & 1 & 1 & 2 \\ 0 & 1 & 1 & K+4 \\ 0 & 0 & -2K & K-K(K+4) \\ 0 & 1 & 0 & 2 \end{pmatrix} \xrightarrow[L_4 \leftarrow L_4 - L_2]{OL_5} \begin{pmatrix} -1 & 1 & 1 & 2 \\ 0 & 1 & 1 & K+4 \\ 0 & 0 & -2K & K-K(K+4) \\ 0 & 0 & -1 & K-2 \end{pmatrix}$$

$$\xrightarrow[L_4 \leftarrow L_4 - \frac{1}{2K}]{OL_6} \begin{pmatrix} -1 & 1 & 1 & 2 \\ 0 & 1 & 1 & K+4 \\ 0 & 0 & -2K & K-K(K+4) \\ 0 & 0 & 0 & -\frac{1}{2}\left(K-K(K+4)\right)+K+2 \end{pmatrix}$$

$$\det(A_K) = -1 \times 1 \times (-2K) \times \left(-\frac{1}{2}\left(K-K(K+4)\right)+K+2\right)$$

$$= 2K \times \left(-\frac{1}{2}\left(K - K^2 - 4K \right) + K + 2 \right)$$

$$= 2K \times \left(\frac{3K + K^2}{2} + K + 2 \right)$$

$$= 2K \times \left(\frac{3K + K^2 + 2K + 4}{2} \right)$$

$$= 2K \left(\frac{K^2 + 5K + 4}{\cancel{2}} \right)$$

$$= K^3 + 5K^2 + 4K$$

$$K^3 + 5K^2 + 4K = 2$$

$$(=) \quad K\left(K^2 + 5K + 4 - \frac{2}{K} \right) = 0$$

$$(=) \quad K = 0 \quad \cup \quad K^2 + 5K + 4 - \frac{2}{K} = 0$$